

DISTRIBUTED SYSTEMS MIDTERM - PART A

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1. Which algorithm is widely used for leader election in distributed consensus?

- A. Paxos
- B. Dijkstra
- C. Bellman-Ford
- D. Floyd-Warshall

2. Consider a distributed key-value store operating under the CAP theorem. During a network partition event, if the

- A. Strong immediate consistency will be sacrificed for eventual consistency.
- B. Latency will decrease to zero across all partitions.

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C. The system will reject all subsequent write requests permanently.

D. Data will be duplicated across all nodes instantaneously.

3. State True or False: In a purely peer-to-peer network, there is no centralized coordinator or server.

A. True

B. False

Answer Key:

1. A

2. A

3. A